

EXTERNAL LONGITUDINAL WEARABLE FOR TIME-SERIES PHYSIOLOGY MONITORING (NON-LIMITING)

A. Technical Field

This section relates to wearable sensing devices configured to collect longitudinal time-series physiology adjacent to external tissue, and to computing systems configured to perform validity gating, artifact rejection, and privacy-preserving metric generation from such signals. The disclosed embodiments further relate to optional correlation of derived metrics with other nodes under consent-gated configurations.

B. Summary (Non-Limiting)

In various embodiments, an external longitudinal wearable is configured to be worn during sleep and/or daily life to collect time-series signals and generate privacy-preserving outputs representing physiologic patterns over time. The wearable may comprise a ring-shaped device, a C-shaped device, or a partially open structure configured for comfort, stable coupling, and repeatable placement. The wearable includes one or more sensing modalities including, without limitation, pressure sensing, strain sensing, displacement sensing, magnetic sensing, optical sensing, temperature sensing, and inertial sensing. A computing system receives signals from the wearable, computes one or more quality indices, selects valid windows, and generates privacy-preserving outputs comprising indices, zones, trend curves, distributions, and confidence indicators.

C. Exemplary Physical Embodiments (Non-Limiting)

In various embodiments, the external wearable comprises a body including an inner contact surface and an outer surface, with electronics arranged within a sealed housing. Exemplary form factors include: 1. a C-shaped wearable comprising two ends separated by a gap, enabling easier placement and removal; 2. a substantially closed ring wearable; and 3. a semi-flexible or rigid body with localized compliant regions configured to improve comfort and contact integrity. In certain embodiments, the inner contact surface is configured as a smooth, continuous interface with soft-touch material, including medical-grade silicone or other biocompatible elastomer. In certain embodiments, the wearable includes internal alignment features, low-profile geometry, and/or compliant pads to support stable coupling and reduce motion-induced artifacts.

D. Sensor Suite (Non-Limiting)

In various embodiments, the external wearable includes one or more sensors including, without limitation: 1. pressure and/or contact sensing, including distributed pressure sensing, capacitive pressure sensing, piezoresistive sensors, and/or force-sensing resistors; 2. strain and/or deformation sensing, including strain gauges, elastomer-embedded sensing elements, and/or film-based strain sensors; 3. displacement sensing, including hall-effect sensors and magnets configured to measure relative displacement and/or geometry changes; 4. optical sensing, including photoplethysmography (PPG) for physiologic confirmation and vascular feature extraction; 5. temperature sensing, including one or more thermal sensors configured to provide context and compensation; and 6. inertial sensing, including a multi-axis IMU configured for motion gating, orientation tracking, and artifact rejection. Any sensor modality may be substituted with an equivalent modality providing substantially similar function. Sensors may be distributed around the wearable body and may be described as sensing regions rather than fixed placements to support design

variation.

E. Collection Modes and Example Use Conditions (Non-Limiting)

In various embodiments, the wearable operates in one or more modes including continuous sampling, periodic sampling with duty cycling, burst sampling during validated windows, and event-triggered sampling based on detected transitions. The wearable may be configured for overnight use, daytime use, or mixed use. The computing system may store time-stamped measurements and associate windows with context tags and quality indices.

F. Standalone Use and Optional Integration Hooks (Non-Limiting)

In standalone embodiments, the wearable generates privacy-preserving outputs for a single user without requiring any other node. In optional multi-node embodiments, derived metrics may be aligned and correlated with other nodes, including intraluminal nodes and/or context nodes, under consent gating and quality-index thresholds. The wearable may provide data to the computing system via authenticated wireless transfer and/or via a docking interface.